Linear Algebra Mastery

Lesson 2: Linear Independence and Basis

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1 Introduction

A set of n-vectors $\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n$ are linearly independent if the only solution to $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n = 0$, and $c_1 = c_2 = \cdots = c_n = 0$ where $c_i \in \mathbb{R}$. If any $c_i \neq 0$, then the vectors are *linearly dependent*. In machine learning, dependent vectors carry redundant information. This causes multicollinearity, rendering the model weights unstable.

A basis is the minimum set of vectors that are linearly independent and span the entire space, meaning the vectors can reach every point in space. The number of vectors in a basis is equal to the TRUE *dimensionality* of a dataset.

2 Formal Definitions

2.1 Linear Independence

Linear independence holds if:

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n = \vec{0}$$

where;

$$c_1 = c_2 = c_3 = \cdots = c_n = 0$$

If $c_i \neq 0$, then the set of vectors is said to be linearly dependent.

Possible Outcomes

1. Linear independence - Each vector add a genuine new information
2. Linear dependence - Redundant vector(s) present, causes multicollinearity.

2.2 Basis

For a set of vectors to be a basis the following two properties must hold:

1. Vectors must be linearly independent
2. Span the entire space. The vectors must reach every point in that space
3. A basis for $\mathbb{R}^n$ has exactly n vectors.

2.3 Standard Basis

There are standard basis in every space, e.g.

$$\text{Standard basis in } \mathbb{R}^2 \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$\text{Standard basis in } \mathbb{R}^3 \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\text{Standard basis in } \mathbb{R}^n \rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

3 Proof: A Set Containing the Zero Vector is Always Linearly Dependent

We want to prove that a set of n-vectors $S = \{\vec{0}, \vec{v}_1, \vec{v}_2, \vec{v}_3, \cdots, \vec{v}_n\}$ is linearly dependent because it contains the zero vector. For a linearly independent set of vectors, the linear combination is $c_0\vec{v}_0 + c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots c_n\vec{v}_n = \vec{0}$.

Let $c_0 = 1$, $v_i \in S$ and $c_i \in \mathbb{R}$

Then;

$$1 \cdot \vec{0} + 0 \cdot \vec{v}_1 + 0 \cdot \vec{v}_2 + \cdots + 0 \cdot \vec{v}_n = \vec{0}$$

The above equation shows that there exist c_i's that are not all zeros. In conclusion, $\{\vec{0}, \vec{v}_1, \cdots, \vec{v}_n\}$ is a linearly dependent set.

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4 Python Results

4.1 Linear Independence Check

Vector $v_1 = (1, 2)$ and $v_2 = (3, 5)$. The rank for the two stacked vectors is 2, meaning that every point in $\mathbb{R}^2$ is reachable, hence linearly independent.

4.2 Linear Dependence Check

Vectors $v_3 = (2, 4)$ and $v_4 = (1, 2)$ are linearly dependent with rank = 1. This means that at least one of the vectors is redundant.

4.3 Standard Basis Verification

The rank for standard basis in $\mathbb{R}^2$ is 2, while in $\mathbb{R}^3$ is 3. This means that with standard basis vectors, you can reach every point in a vector space.

4.4 ML Connection — Feature Matrix

The feature matrix $X = \begin{pmatrix} 1 & 2 \\ 3 & 5 \\ 2 & 4 \end{pmatrix}$ have a rank = 2 when checking among features. With 2 features present, it means that the columns are not redundant. A matrix has one rank, however redundancy is interpreted differently depending on whether you are comparing rank against the number of rows(entries) or columns(features).

5 ML Connection

Before training a model, check for redundant features. Redundant features add no new information causing multicollinearity, which makes a model weights to be unstable.

6 Reflection

The hardest concept today was to proof that a set of vectors containing the zero vector is always dependent. I was surprised by how you can check redundancy for features or rows.